



Countering Multi-modal Representation Collapse through Rank-targeted Fusion

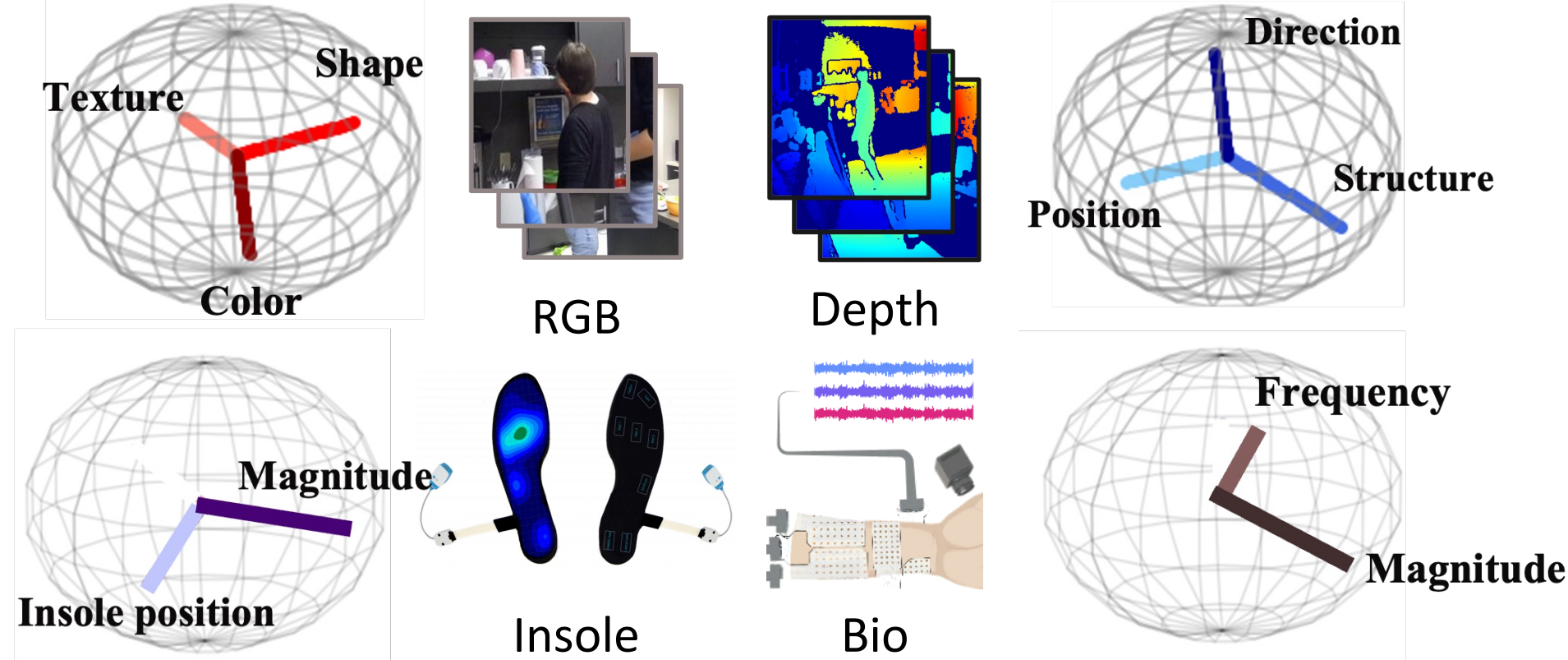
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Introduction

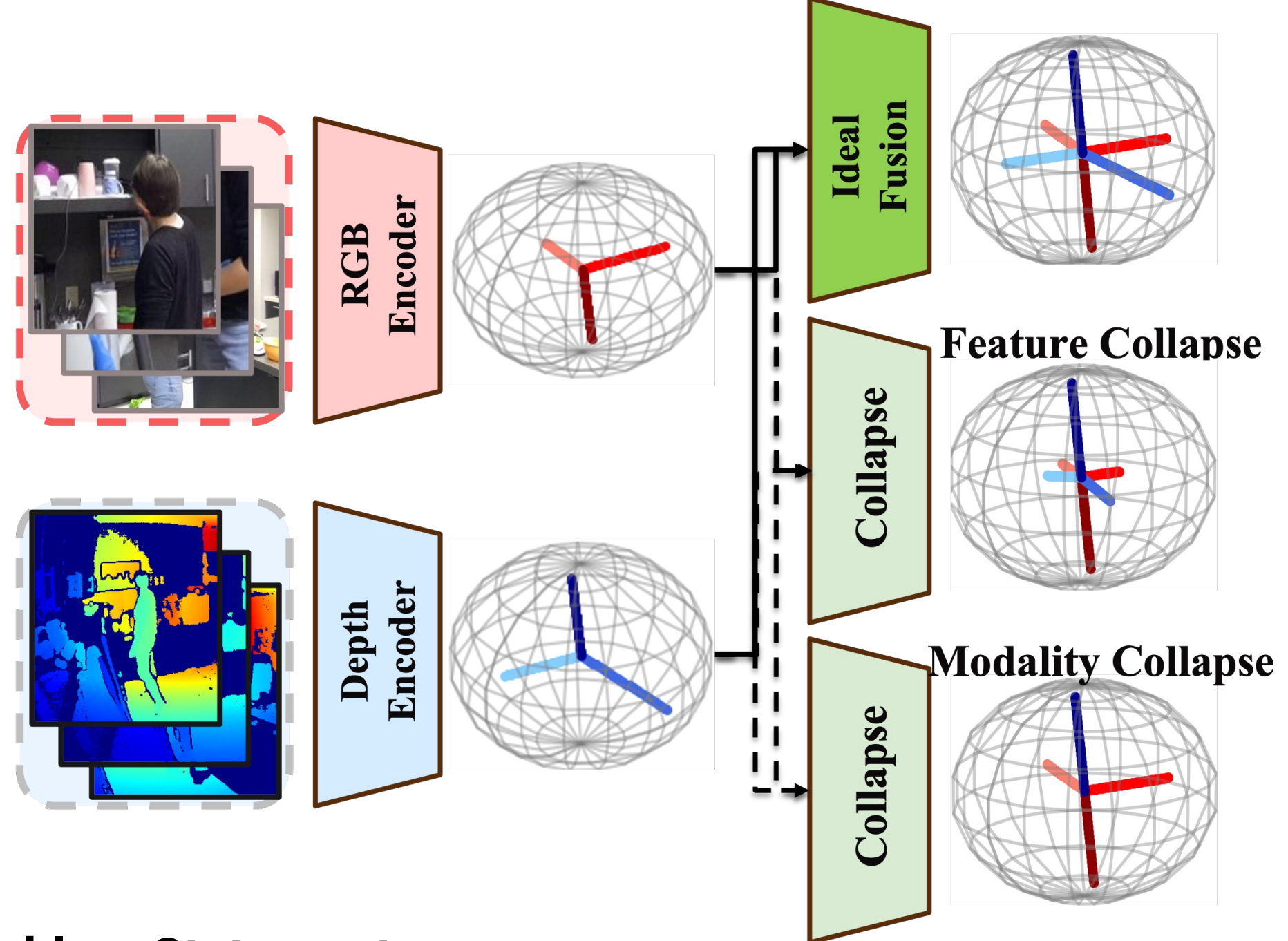
What is Multi-modal Learning?

- ❑ We understand the world from diverse source of modalities.
- ❑ Each modality complements other modalities.



Challenges in Multi-modal Learning

- ❑ **Feature Collapse**: Preserve dominant features only
- ❑ **Modality Collapse**: Ignore information from certain modality



Problem Statement

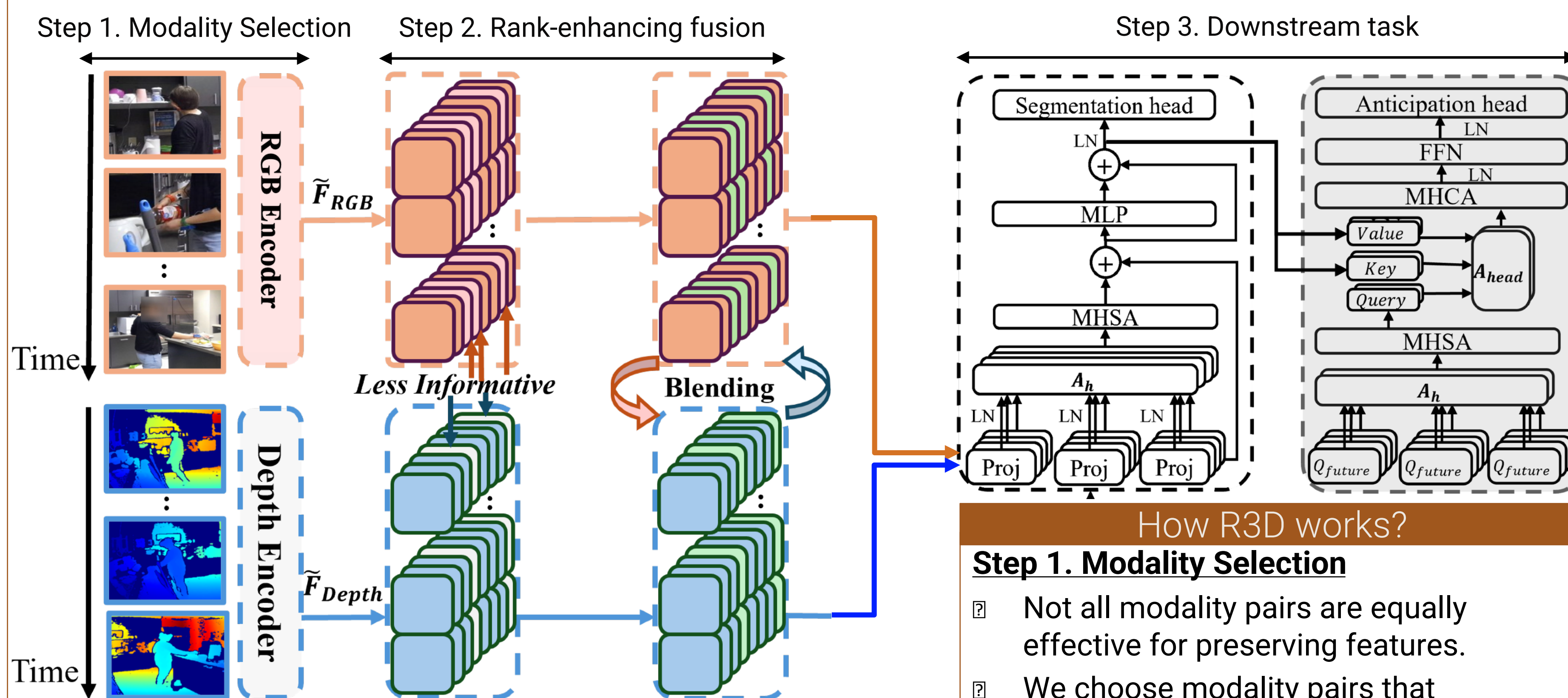
- ❑ How do we design multi-modal learning method to encourage leveraging the full information?
- ❑ What is the best combination of modality pairs to counter feature / modality collapse in multi-modal learning?

Information Theory Lens

Effective Rank

- ❑ $ERank(Z) = \exp(-\sum \frac{\sigma_k}{\sum \sigma_j} \log(\frac{\sigma_k}{\sum \sigma_j}))$, where σ_k as the singular values of feature matrix Z .

R3D Architecture

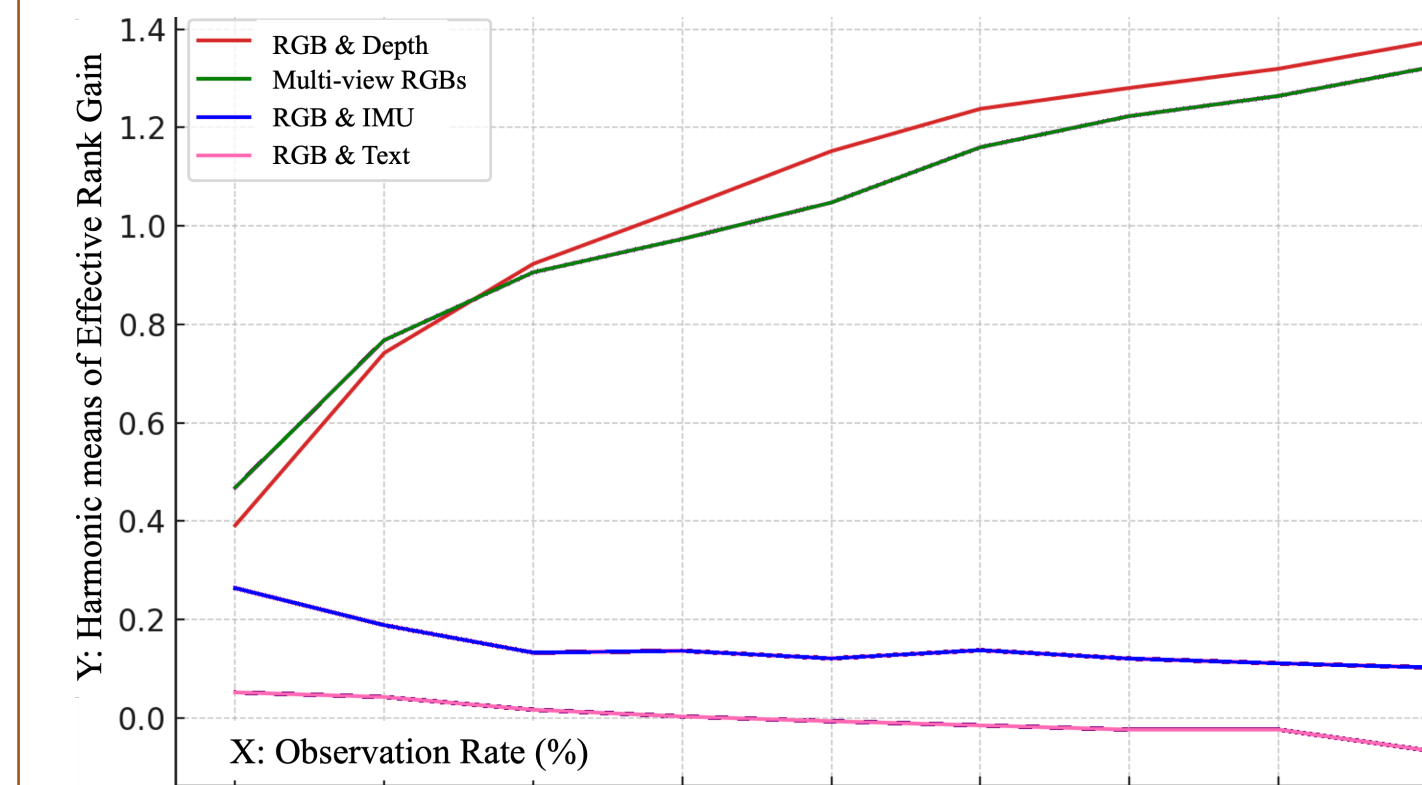


How R3D works?

Step 1. Modality Selection

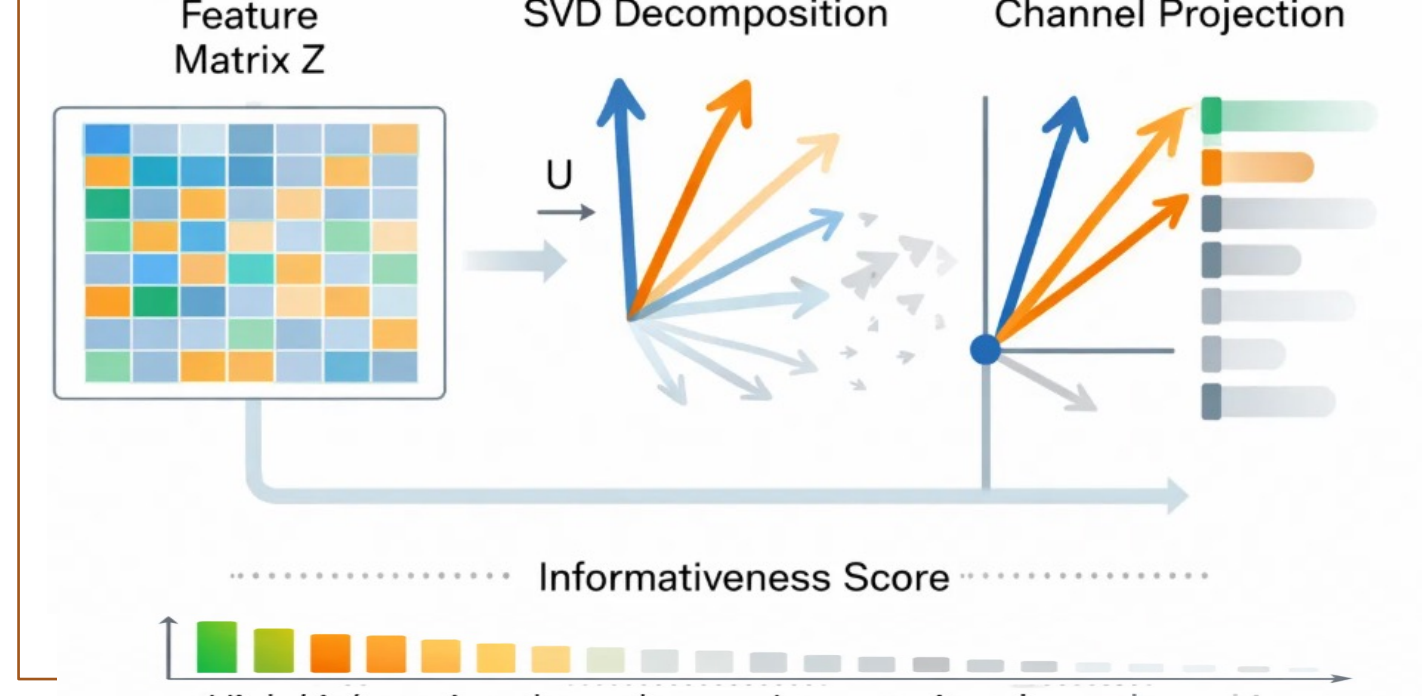
- ❑ Not all modality pairs are equally effective for preserving features.
- ❑ We choose modality pairs that maximize **mutual** feature diversity gain.

$$\frac{2 * \Delta_{RGB} * \Delta_{Depth}}{\Delta_{RGB} + \Delta_{Depth}} > \frac{2 * \Delta_{RGB} * \Delta_{IMU}}{\Delta_{RGB} + \Delta_{IMU}}, \frac{2 * \Delta_{RGB} * \Delta_{Text}}{\Delta_{RGB} + \Delta_{Text}}$$

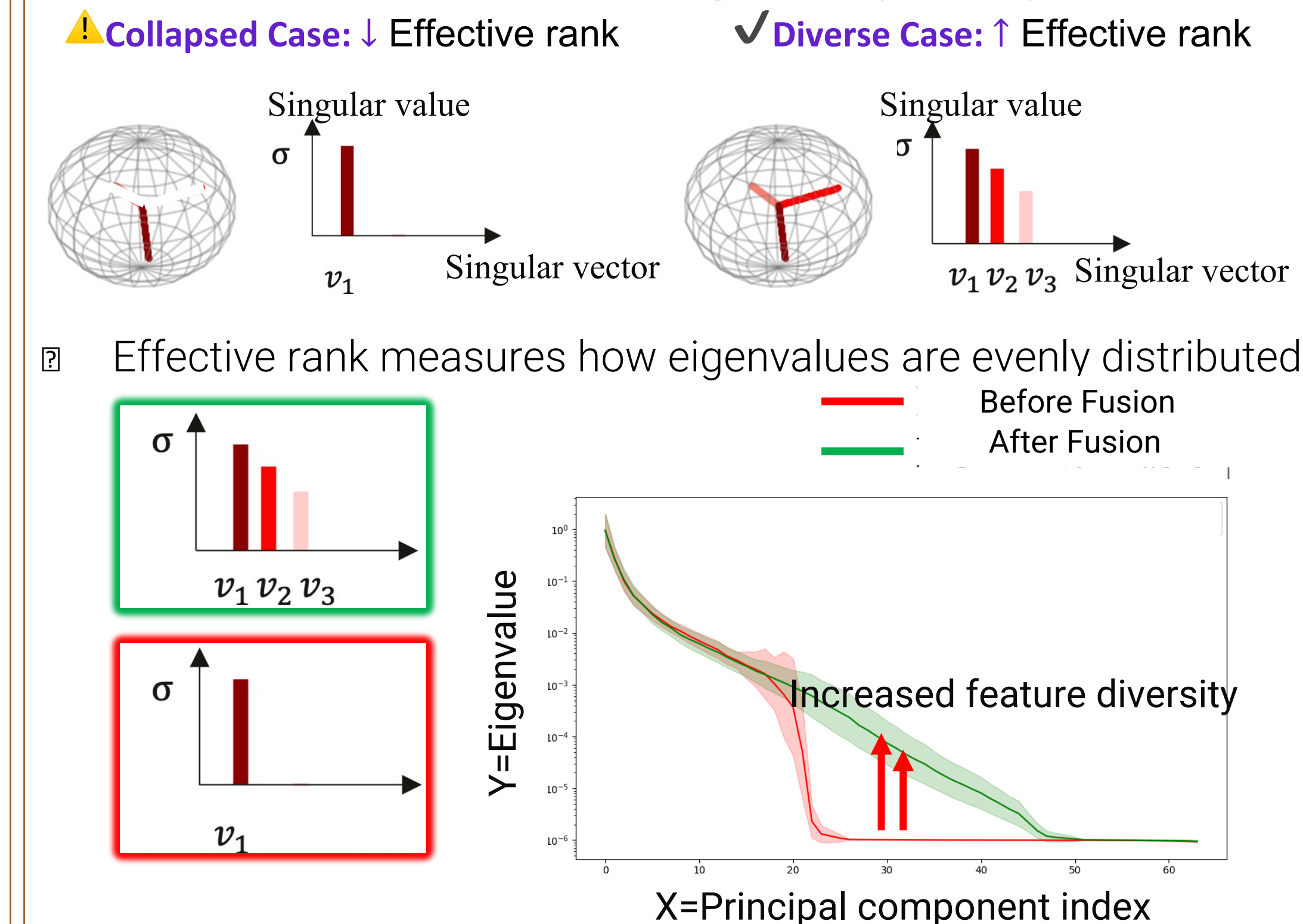


Step 2. Rank-enhancing Fusion

- ❑ R3D selectively targets the less informative channels.



Information Theory Lens (Cont'd)



What is an 'Informative' representation space?

- ❑ We define matrix A is more informative than matrix B , when matrix A represents more diverse features than matrix B .
- ❑ Hence, matrix A is more informative than matrix B when $ERank(A) > ERank(B)$

Experiments / Results

State-of-the-art Comparison

- ❑ Action Anticipation downstream task
- ❑ Tested on different α (Observation rate), β (prediction rate)

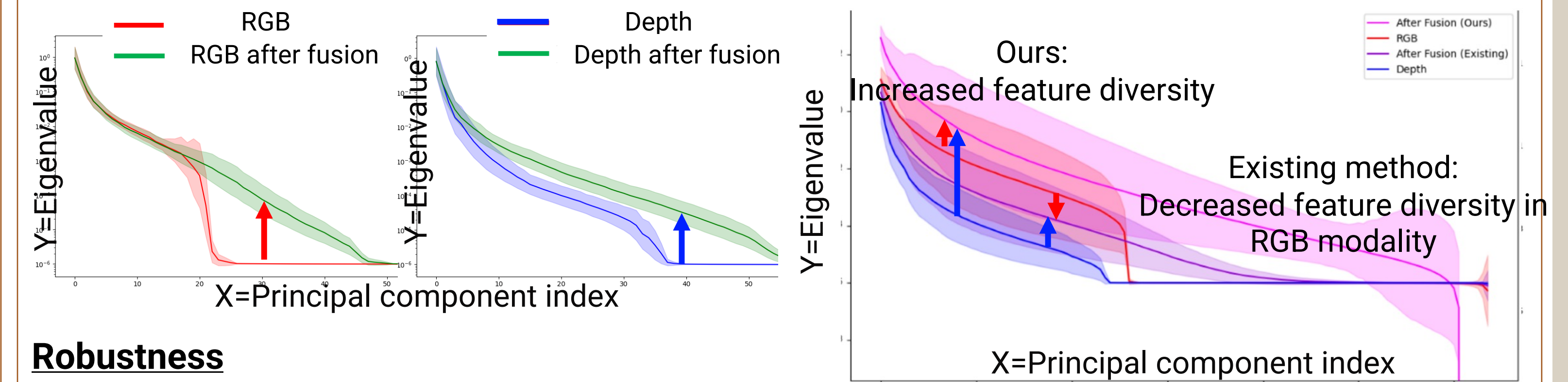
Dataset	Input	Methods	$\beta(\alpha = 0.2)$				$\beta(\alpha = 0.3)$			
			0.1	0.2	0.3	0.5	0.1	0.2	0.3	0.5
DARai [32] (Coarse)	Uni	RNN [1]	22.40	22.59	20.71	18.38	30.75	25.34	25.99	23.17
		CNN [1]	13.28	13.70	12.99	13.90	19.85	16.68	18.69	17.48
		FUTR [22]	25.05	25.11	24.48	23.18	40.71	33.57	33.43	30.79
		GTAN [72]	27.70	29.13	27.35	26.11	42.47	42.45	42.27	34.37
		R3D (Ours)	31.52	30.52	28.48	26.66	44.01	37.10	37.58	34.44
Multi		AFPT [76]	23.14	24.78	23.62	21.02	33.82	29.25	28.33	25.45
		m&cm-Ant [36]	25.75	25.70	24.24	23.12	42.00	34.71	34.49	31.34
		R3D (Ours)	33.44	32.14	30.56	29.59	50.29	42.05	43.41	40.25
	Uni	RNN [1]	7.45	5.69	4.66	3.58	9.07	7.35	6.40	5.04
		CNN [1]	7.70	6.28	5.38	3.98	6.69	5.48	5.10	3.86
DARai [32] (Fine-grained)		FUTR [22]	17.09	12.34	10.39	8.18	18.59	15.83	13.26	10.81
		GTAN [72]	25.66	24.06	22.60	22.05	26.30	25.43	22.40	18.88
		R3D (Ours)	20.12	18.49	16.11	13.28	24.62	22.93	22.78	18.68
	Multi	AFPT [76]	14.13	13.38	13.49	9.53	17.01	17.26	15.18	10.91
		R3D (Ours)	32.57	25.92	24.02	16.68	26.97	28.43	24.81	18.02

Action Segmentation downstream task

Dataset	Methods	Frame-wise metrics		Segment-wise metrics	
		Accuracy		Edit score	
DARai [32] (Coarse)	ActFusion [23]	31.38		26.62	
	R3D (Ours)	33.32		31.21	
DARai [32] (Fine-grained)	ActFusion [23]	17.83		18.16	
	R3D (Ours)	20.79		28.98	
NTURGBD [59]	ActFusion [23]	13.26		6.00	
	R3D (Ours)	13.26		8.93	

Feature Diversity

- ❑ R3D shows improved feature diversity in both modalities



Robustness

- ❑ Tested on various noise perturbations (clean, small noise, big noise).
- ❑ R3D adaptively relies more on cleaner modalities, showing robustness on big noise setting.

